0147

National University of Computer and Emerging Sciences

FAST School of Computing Spring-2023 Islamabad Campus

CS-1004: Object Oriented Programming (CS)

Serial No:

Sessional Exam-II

Total Time: 1 Hour

Total Marks: 60

Signature of Invigilator

Monday, 10th April, 2023

Course Instructors

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Aneeq Ahmed M.Ak 23i-1167 CS-B

Student Name Roll No. Section Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have **ten (10)** different printed pages including this title page. There are a total of **2** questions. And write your instructor's complete office number in front of point 6 to secure bonus marks.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

C-205F

	Q-1	Q-2	Total
Marks Obtained	30	20	50
Total Marks	30	30	60

+5 = 55

V. good

Question 01 [30 marks]

What would be the output produced by executing the following C++ codes? Identify errors, if any (either write output or error (syntax/runtime), both will not be accepted). All the code snippet contains #include<iostream> and using namespace std;

A [3 Marks]

```
class Box{
  int capacity;
bool operator<(Box b){
        return this->capa city < b.capacity ? true : false;}
public:
  Box(){}
  Box(double capacity){ this->capacity = capacity; }
};
int main(){
        Box b1(10);
        Box b2 (14);
        if(b1 < b2){
                cout<<"Box 2 has large capacity.";
        }
        else{
                cout<<"Box 1 has large capacity.";
        }
        return 0;
}
```

Output/Error:

Error

bcz overloaded(<) is private, so not accessable inside main().

3

B [5 Marks]

```
class Point{
   int x, y;
   public:
   Point(int x=0, int y=0){
      this->x=x;      Point::y=y;
      (*this)();
   }
   void operator()(){
     cout<<" ("<<x<<","<<y<<") " <<endl;
   }
   Point& operator()(int y){
     this->y=y;
     return *this;
   }
   ~Point(){
     cout<<"Point is going";   (*this)();
     }
  }p3;
  int main() {
   Point *p=new Point(5,6);
    static Point p1(p3);
   p1(9)(8);
   delete p;
    Point p2(7);
   cout<<"----------"<<endl;
return 0;
}
```

Output/Error:

(0, 0)
(5, 6)
Point is going (5, 6)
(7, 0)

Point is going (7, 0)
Point is going (0, 8)
Point is going (0, 0)

5

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C [7 Marks]

```
class Complex{
      double r, i;
   public:
Complex(double r=1.0,double i=1.0){
   set(r,i);
}
void set(double r,double i){
    Complex::r = r; this->i = i;
}
void print(){
   if (i<0)
   cout << r << "" << i <<"i"<<endl;
   else
   cout << r << "+" << i <<"i"<<endl;
}
Complex operator+(Complex R){
    Complex tmp;
    tmp.r = r + R.r;
    tmp.i = i + R.i;
    return tmp;
}                                  post fuction
Complex operator++(){
    Complex tmp=*this;
    r++; i++;
    return tmp;
}
Complex operator++(int){          pre fun
   ++(*this);
   return *this;
}
};
int main()
{
Complex A(3,4), B(5,-6);
A.print();
B.print();
Complex C;
C= A+B;
C.print();
(++A).print();
C = ++A;
C.print();
(A++).print();
A.print();
   return 0;
}
```

Output/Error:

3+4i
5-6i
8-2i
3+4i
4+5i
6+7i
6+7i

7

D [5 Marks]

```
class Mystery {
public:
static int n;
Mystery() {cout << n++<<endl; }
Mystery(int i) {n=i;cout << n<<endl;}
static void somefunc(){  n=5;}

Mystery(Mystery const& otherNum){
   n+=5;
   cout << n<<endl;
}
 ~Mystery(){cout<< --n <<" ";}
 }a;

 void fun(Mystery n){
    cout<<n.n<<endl;
    n.somefunc();
 }
 int Mystery::n=0;

  int main(){
 Mystery b(9), c;
 fun(b);
 return 0;
  }
```

Output/Error:

0
9
9
15
15
4 3 2 1

5

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E [10 Marks]

```
class ZooCage
{
   int cageNumber;
   ZooCage* link;
public:
   ZooCage(int n):cageNumber(n),link(NULL){}
   int getCageNumber() {return cageNumber;}
   ZooCage*& getLink() {return link;}
}*start=NULL;
void fun(ZooCage *& H, int num)
{
if(H){
     fun(H->getLink(),num);
     return;
     }
   H = new ZooCage(num);
}
void fun(ZooCage * H)
{
if(H){
  fun(H->getLink());
  cout<<H->getCageNumber()<<endl;
}
}
int main(){
  fun(start,4);
  fun(start,2);
  ZooCage * temp=new ZooCage(5);
  temp->getLink()=start->getLink();
  start->getLink()=temp;
  fun(start,3);
  fun(start);
return 0;
}
```

Output/Error:

3
2
5
4

10

Question 02 [30 marks]

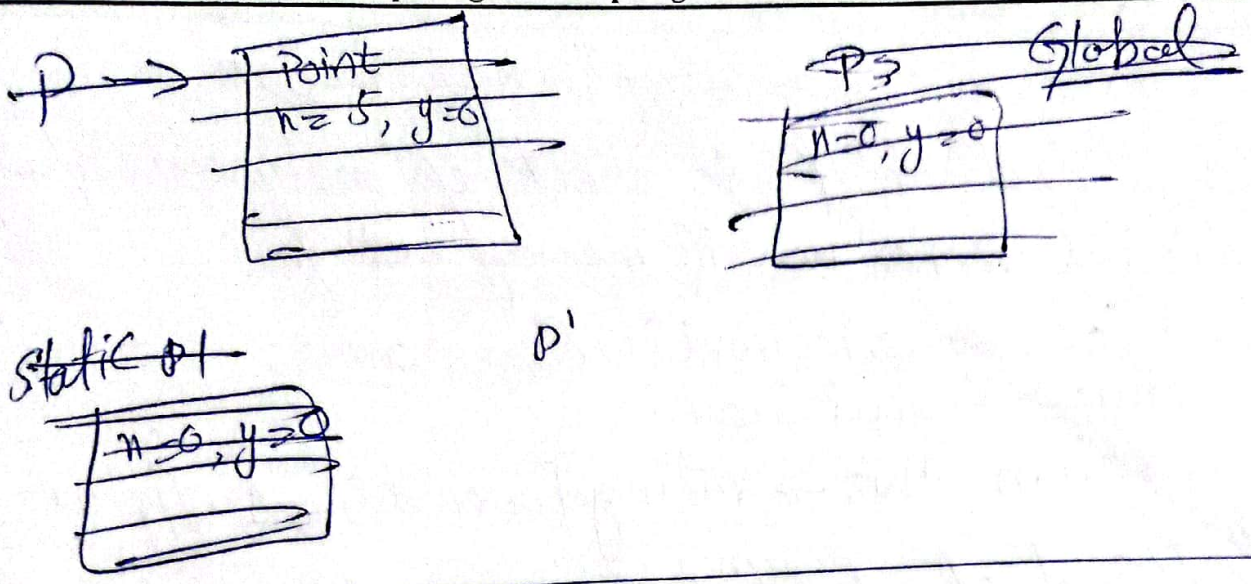
Your goal is to write a program for creating a MovieStore. Your MovieStore should allow for storage of many movie names, there **(unique)** IDs and ratings. Your MovieStore should allow facility of performing following operations, (minimum) data members required are given (you can add more). You need to add the necessary functions as per sequence, following the sequence will give you 2 marks:

```
int main(){
        MovieStore s1, s2(10); // create two stores of size 5 and 10

        s1["Hobbit"]= 4.5;
/*  If previously not added, add a movie with average rating of 4.5
    If already exist , overwrite rating by 4.5
    If memory finished print "Memory finished overwriting rating of last movie"
    And Update accordingly */

        s1["Lord of the Rings"] = 5;

        cout << s1[2]<<endl;
/*  get rating for movie id 2 (Lord of the Rings => 5).
    If id not found print "id not found"
    And return -1 rating.*/

        s2=s1; // copies elements from s1 to s2

        s1["Lion King"] = 8;
        cout << s2; // should display all the Movie store information
return 0;
}
```

Expected Output:

```
5
MovieStore:
 Id    Movie              Rating
 1     Hobbit             4.5
 2     Lord of the Rings  5
```

```
class MovieStore{
    string * names;
    int * ids;
    float * rating;
    int size;
    int currentSize;
.....
```




```
public:
MovieStore (int size=5)
{ this->size=size;  this->currentSize=0;
  this->names = new string[size];
  this->ids = new int[size];
  this->rating = new float[size];
}
float & operator[](string name)
{ // bool flag=false;
  for(int i=0; i<this->currentSize; i++)
  {
    if(this->names[i] == name)
    {
      return this->rating[i];
    }
  }
  ~~this->names[currentSize] = name;~~
  if(this->currentSize == this->size-1)
  {
    cout<<"Memory finished overwriting
          rating of last movie";
  this->names[currentSize]=name;
```

3

5

12
8
20

```
    return this->rating[currentSize];
} //End of if to check if memory is full
//Control reaches here if memory not full
    this->names[currentSize] = name;
    this->currentSize++;
    return this->rating[currentSize-1]; //return currentSize-1
} //End of function

float operator[](int number)
{
    if(this->currentSize+1 <= number)
    {
        cout<<"id not found";
        return -1;
    } //if id exceeds currentSize
    return this->rating[number-1];
}
MoviesStore(const &MoviesStore M)
{
    this->size = M.size;
    this->currentSize = M.currentSize;
    for(int i=0; i<
    this->names = new string[this->size];
    this->ids = new int[this->size];
```

```
this->rating = new float[this->size];
for(int i=0; i<this->currentSize; i++)
{
    this->names[i] = M.names[i];
    this->rating[i] = M.rating[i];
    this->ids[i] = M.ids[i];
}
}// End of function
~MovieStore()
{
    delete[] names;
    delete[] ids;
    delete[] ratings;
}
MoviesStore operator=(const & MovieStore M)
{ this->size = M.size; this->CurrentSize = M.CurrSize
  this->names = new string[size];
  this->ids = new int[size]
  this->rating = new float[size];
  for(int i=0, i<currSize; i++)
  {
      this->name[i] = M.name;
      this->rating[i] = M.rating;
      this->id[i] = M.ids;
  }
};
```

3